5.1.1. Stacked Plot

Problem Statement:

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your tasks are:
- Create a stacked area plot using the data.
- Label the x-axis as 'Month', the y-axis as 'Temperature', and provide the title 'Temperature Variation' for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Code:

```
import matplotlib.pyplot as plt
import pandas as pd

data = {
    'Month': ['January','February','March','April','May','June',
              'July','August','September','October','November','December'],
    'City_A_Temperature': [5,7,10,14,18,22,25,24,20,15,10,6],
    'City_B_Temperature': [2,3,7,10,14,18,21,20,16,11,6,3],
    'City_C_Temperature': [0,1,4,8,12,16,19,18,14,9,4,1]
}
df = pd.DataFrame(data)
plt.stackplot(df['Month'], df['City_A_Temperature'],
             df['City_B_Temperature'], df['City_C_Temperature'])
plt.xlabel('Month')
plt.ylabel('Temperature')
plt.title('Temperature Variation')
plt.legend('upper left')
plt.show()
```

Output:

Explorer

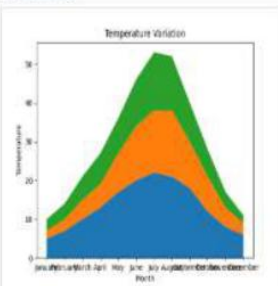
Average time
0.568 s
568.00 ms

Maximum time
0.568 s
568.00 ms

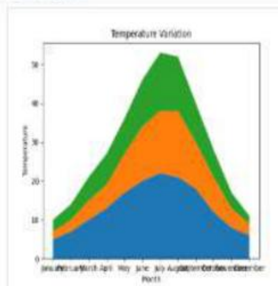
1 out of 1 shown test case(s) passed

Test case 1 568 ms Debug

Expected output



Actual output



Debugger

5.2.1 Titanic Dataset

Problem Statement:

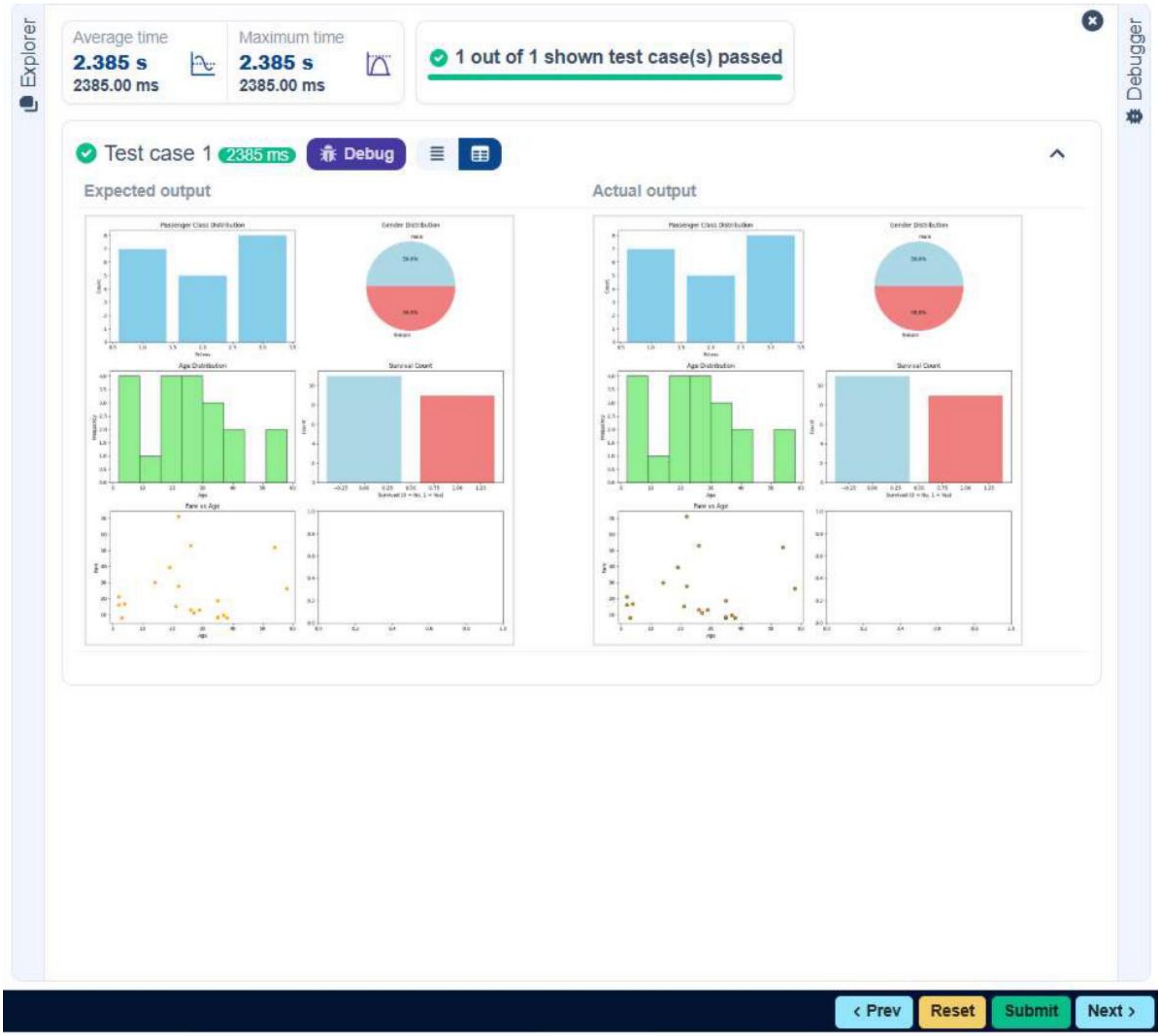
Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions.

Visualizations: Represent these trends using Matplotlib in a 3x2 grid (3 rows and 2 columns).
- Bar Plot (Pclass Distribution): Distribution of passengers across passenger classes (skyblue).
- Pie Chart (Gender Distribution): Distribution of male and female passengers (lightgreen / lightcoral).
- Histogram (Age Distribution): Distribution of passenger ages, 10 bins, black edges.
- Bar Plot (Survival Count): Count of survived vs not survived (lightblue / lightcoral).
- Scatter Plot (Fare vs Age): Relationship between Fare and Age of passengers (orange).

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('titanic.csv')
fig, axes = plt.subplots(3, 2, figsize=(10, 10))

axes[0,0].bar(df['Pclass'].value_counts().index,
             df['Pclass'].value_counts(), color='skyblue')
axes[0,0].set_title('Passenger Class Distribution')
axes[0,0].set_xlabel('Pclass')
axes[0,0].set_ylabel('Count')

axes[0,1].pie(df['Gender'].value_counts(),
             labels=df['Gender'].value_counts().index,
             autopct='%1.1f%%', colors=['lightgreen','lightcoral'])
axes[0,1].set_title('Gender Distribution')

axes[1,0].hist(df['Age'].dropna(), bins=10, edgecolor='black')
axes[1,0].set_title('Age Distribution')
axes[1,0].set_xlabel('Age')
axes[1,0].set_ylabel('Frequency')

axes[1,1].bar(df['Survived'].value_counts().index,
             df['Survived'].value_counts(),
             color=['lightblue','lightcoral'])
axes[1,1].set_xlabel('Survived (0 = No, 1 = Yes)')
axes[1,1].set_ylabel('Survival Count')

axes[2,0].scatter(df['Age'], df['Fare'], color='orange')
axes[2,0].set_title('Fare vs Age')
axes[2,0].set_xlabel('Age')
axes[2,0].set_ylabel('Fare')

plt.tight_layout()
plt.show()
```

Output:



Explorer
Average time
2.385 s
2385.00 ms
Maximum time
2.385 s
2385.00 ms
1 out of 1 shown test case(s) passed
Debugger
Test case 1
2385 ms
Debug
Expected output
Actual output
< Prev
Reset
Submit
Next >

5.2.2 Histogram of Passenger Information of Titanic

Problem Statement:

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram.
2. Set the edge color of the bars to black (k).
3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.
4. Add the title 'Age Distribution' to the histogram.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.hist(data['Age'], bins=30, edgecolor='k')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age Distribution')
plt.show()
```

Output:

Average time
0.647 s
647.00 ms

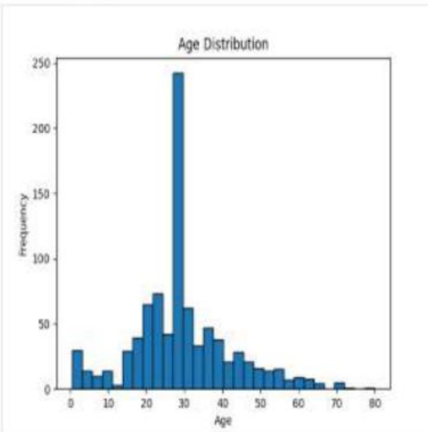
Maximum time
0.647 s
647.00 ms

1 out of 1 shown test case(s) passed

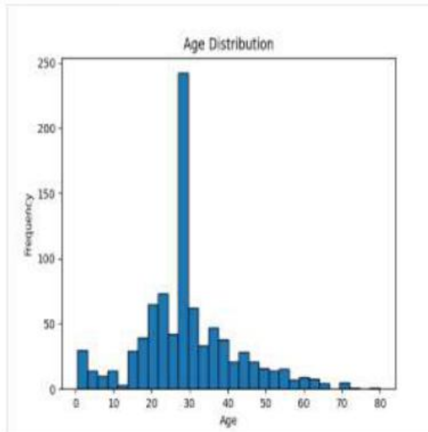
Debugger

Test case 1 647 ms Debug

Expected output



Actual output



‹ Prev Reset Submit Next ›

5.2.3 Bar Plot of Survival Rate of Passengers

Problem Statement:

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. Specifications:

1. Use the 'Survived' column to show count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title 'Survival Count' to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

survival_counts = data['Survived'].value_counts()
survival_counts.plot(kind='bar')
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.show()
```

Output:

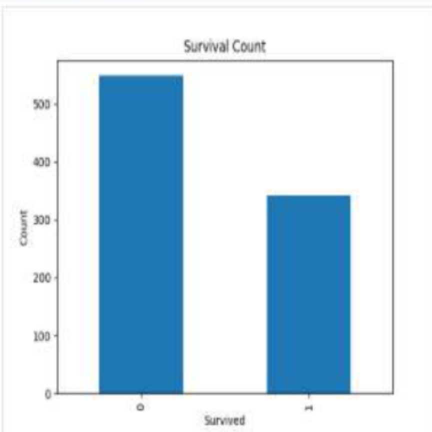
Average time
0.647 s
647.00 ms

Maximum time
0.647 s
647.00 ms

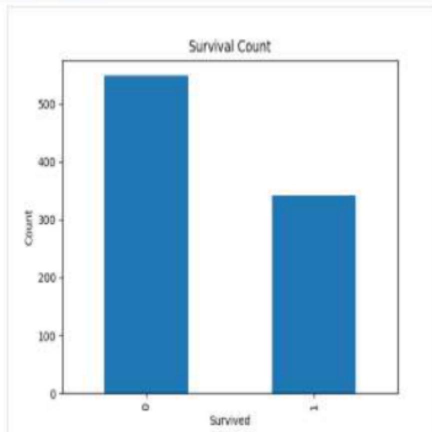
1 out of 1 shown test case(s) passed

Test case 1 647 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.4 Bar Plot for Survival by Gender

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender. Specifications:

1. Group the data by the 'Sex' column, then use value_counts() to count occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a stacked bar chart to display the survival counts.
3. Add the title 'Survival by Gender' to the chart.
4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

survival_by_gender =
data.groupby('Sex')['Survived'].value_counts().unstack().fillna(0)
survival_by_gender.columns = ['Not Survived', 'Survived']
survival_by_gender.plot(kind='bar', stacked=True)
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(title=None)
plt.show()
```

Output:

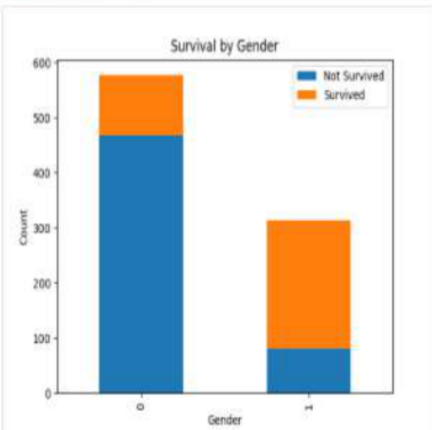
Average time
0.774 s
774.00 ms

Maximum time
0.774 s
774.00 ms

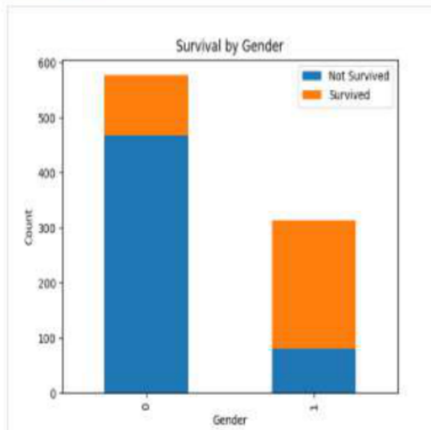
1 out of 1 shown test case(s) passed

Test case 1 774 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.5 Bar Plot for Survival by Pclass

Problem Statement:

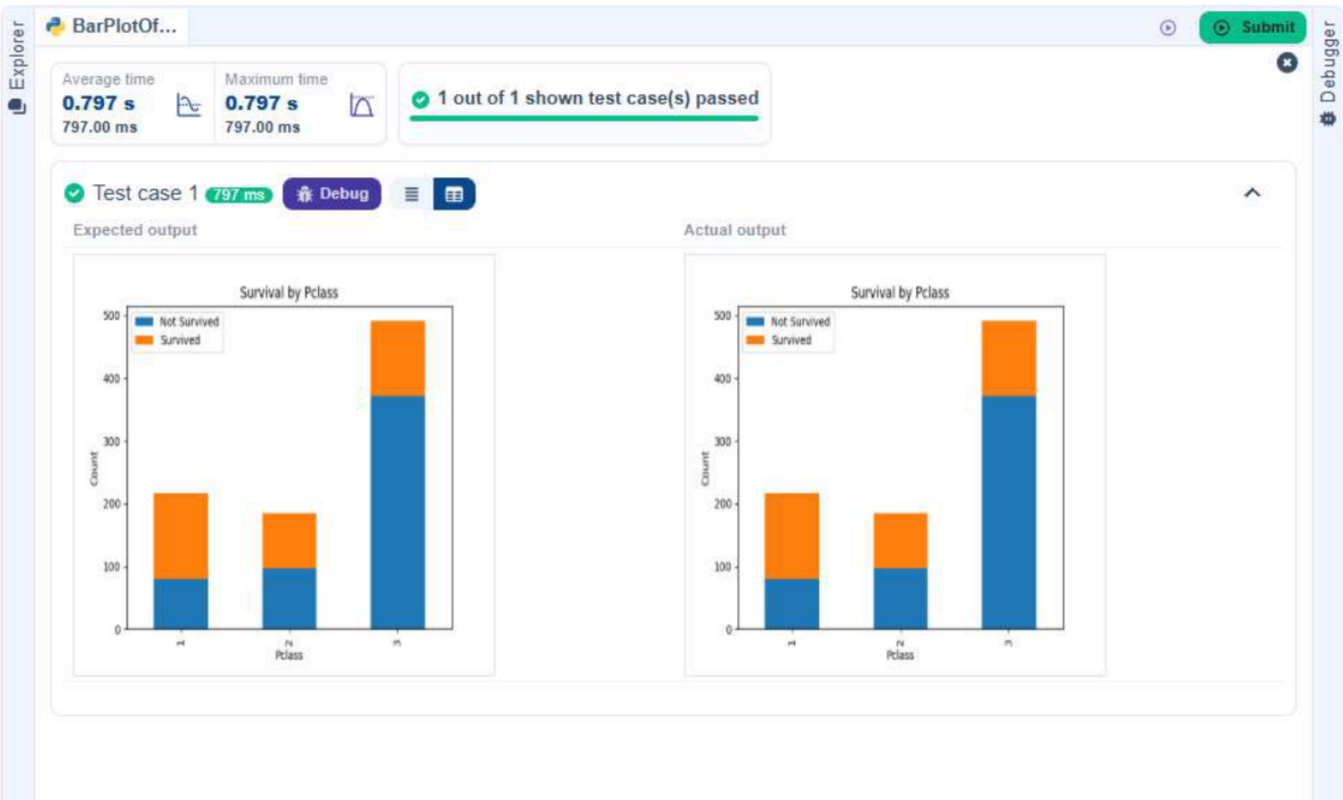
Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass). Specifications:

1. Group the data by Pclass column and count survivors for each class using value_counts().
2. Use a stacked bar chart to display the survival counts.
3. Add the title 'Survival by Pclass' to the chart.
4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.
5. The legend should indicate 'Not Survived' and 'Survived'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

survival_by_class =
data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)
survival_by_class.columns = ['Not Survived', 'Survived']
survival_by_class.plot(kind='bar', stacked=True)
plt.title('Survival by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Count')
plt.legend(title=None)
plt.show()
```

Output:


BarPlotOf...
Submit
Explorer
Debugger
Average time
0.797 s
797.00 ms
Maximum time
0.797 s
797.00 ms
1 out of 1 shown test case(s) passed
Test case 1
797 ms
Debug
Expected output
Actual output
Survival by Pclass
Not Survived
Survived
Count
Pclass
Survival by Pclass
Not Survived
Survived
Count
Pclass

5.2.6 Bar Plot for Survival by Embarked

Problem Statement:

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. Specifications:

1. Use the Embarked column to determine the embarkation location. After converting to dummy variables using pd.get_dummies(), plot the survival count based on the Embarked_Q column in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title 'Survival by Embarked' to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

grouped = data.groupby('Embarked_Q')['Survived'].value_counts().unstack().fillna(0)
grouped.columns = ['Not Survived', 'Survived']
grouped.plot(kind='bar', stacked=True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.legend(title=None)
plt.show()
```

Output:

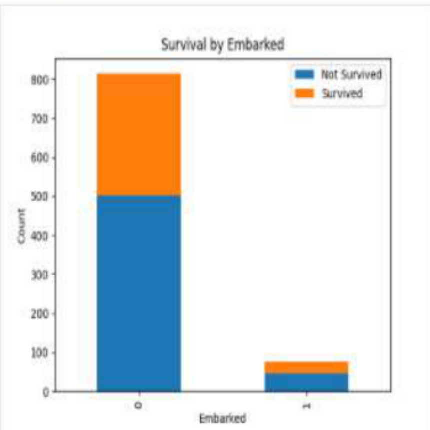
Average time
0.827 s
827.00 ms

Maximum time
0.827 s
827.00 ms

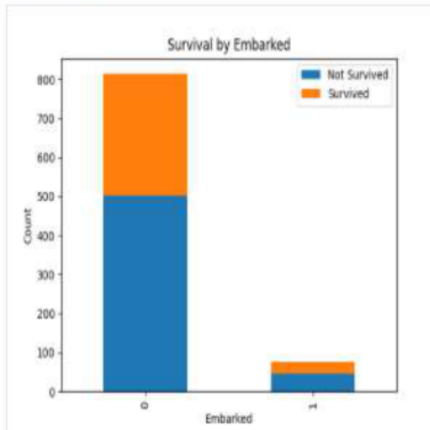
1 out of 1 shown test case(s) passed

Test case 1 827 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.7 Box Plot for Age Distribution

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. Specifications:

1. Use the Pclass column to group the data for the boxplot.
2. Set the title of the plot to 'Age by Pclass'.
3. Remove the default subtitle with plt.suptitle('').
4. Label the x-axis as 'Pclass' and the y-axis as 'Age'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.figure(figsize=(8, 6))
data.boxplot(column='Age', by='Pclass')
plt.suptitle('')
plt.title('Age by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.show()
```

Output:

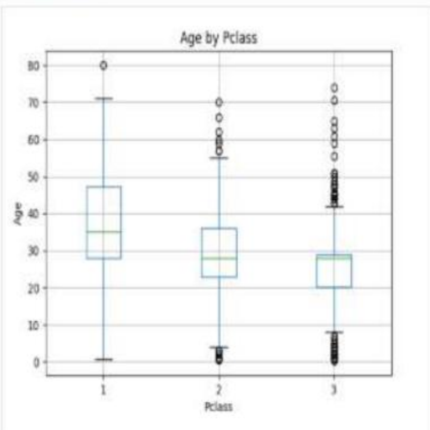
Average time
0.704 s
704.00 ms

Maximum time
0.704 s
704.00 ms

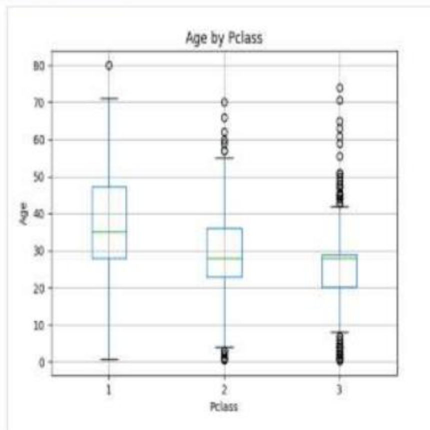
1 out of 1 shown test case(s) passed

Test case 1 704 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.8 Box Plot for Age by Survival

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. Specifications:

1. Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to 'Age by Survival'.
3. Remove the default subtitle with plt.suptitle('').
4. Label the x-axis as 'Survived' and the y-axis as 'Age'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.figure(figsize=(8, 6))
data.boxplot(column='Age', by='Survived')
plt.suptitle('')
plt.title('Age by Survival')
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()
```

Output:

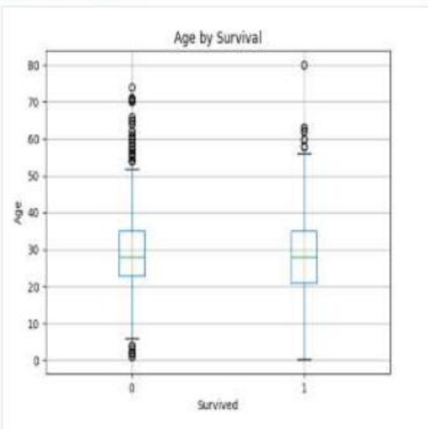
Average time
0.674 s
674.00 ms

Maximum time
0.674 s
674.00 ms

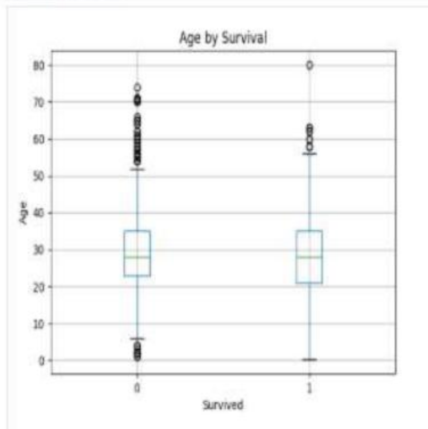
1 out of 1 shown test case(s) passed

Test case 1 674 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.9 Box Plot for Fare by Pclass

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the Pclass column to group the data for the boxplot.
2. Set the title of the plot to 'Fare by Pclass'.
3. Remove the default subtitle with plt.suptitle('').
4. Label the x-axis as 'Pclass' and the y-axis as 'Fare'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.figure(figsize=(8, 6))
data.boxplot(column='Fare', by='Pclass')
plt.suptitle('')
plt.title('Fare by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Fare')
plt.show()
```

Output:

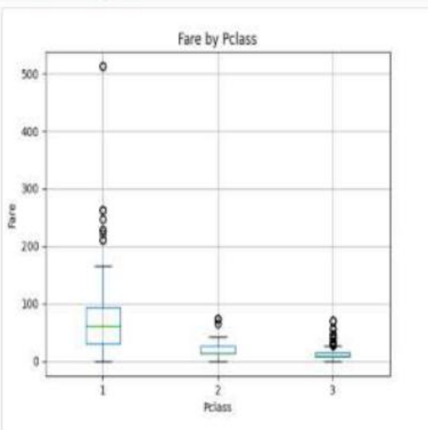
Average time
0.794 s
794.00 ms

Maximum time
0.794 s
794.00 ms

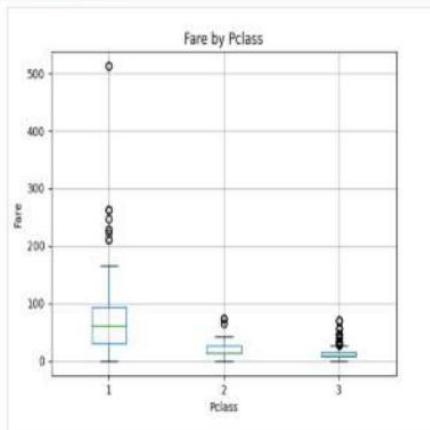
1 out of 1 shown test case(s) passed

Test case 1 794 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.10 Scatter Plot for Age vs. Fare

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the Age column for the x-axis and the Fare column for the y-axis.
2. Set the title of the plot to 'Age vs. Fare'.
3. Label the x-axis as 'Age' and the y-axis as 'Fare'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.figure()
plt.scatter(data['Age'], data['Fare'])
plt.title('Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```

Output:

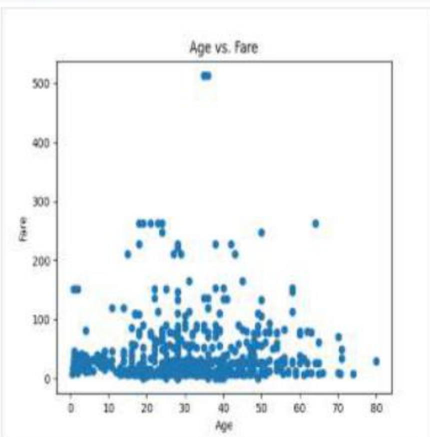
Average time
0.602 s
602.00 ms

Maximum time
0.602 s
602.00 ms

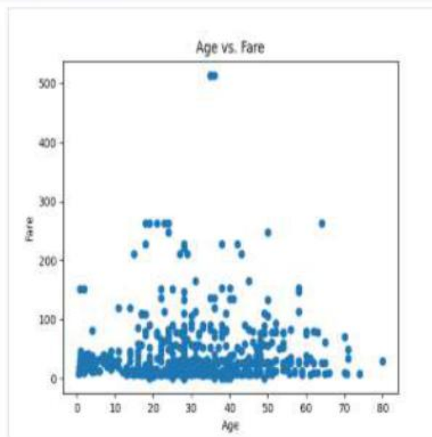
1 out of 1 shown test case(s) passed

Test case 1 602 ms Debug

Expected output



Actual output



Terminal Test cases

‹ Prev Reset Submit Next ›

5.2.11 Scatter Plot for Age vs. Fare by Survival

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the Age column for the x-axis and the Fare column for the y-axis.
2. Color the points based on the Survived column. Red for passengers who did not survive (Survived = 0). Blue for passengers who survived (Survived = 1).
3. Set the title of the plot to 'Age vs. Fare by Survival'.
4. Label the x-axis as 'Age' and the y-axis as 'Fare'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('Titanic-Dataset.csv')
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

plt.figure()
colors = {0: 'red', 1: 'blue'}
plt.scatter(data['Age'], data['Fare'],
            c=data['Survived'].apply(lambda x: colors[x]))
plt.title('Age vs. Fare by Survival')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```

Output:

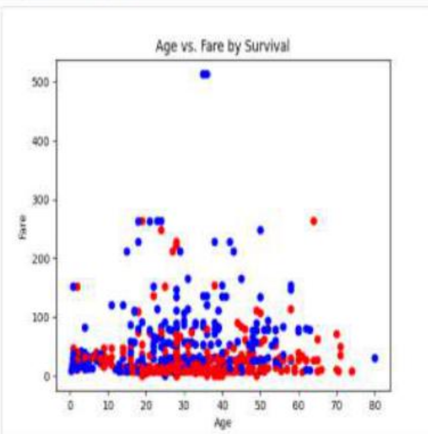
Average time
0.671 s
671.00 ms

Maximum time
0.671 s
671.00 ms

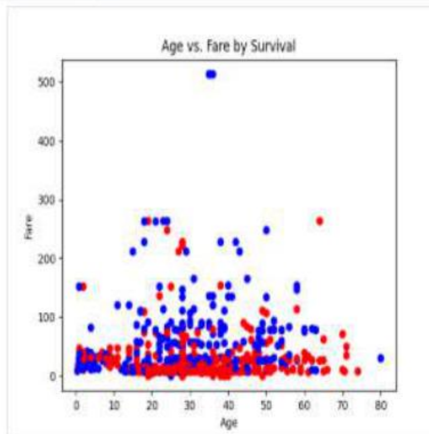
1 out of 1 shown test case(s) passed

Test case 1 671 ms Debug

Expected output



Actual output



Terminal Test cases

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